7 (a) (i)

$$E = \frac{hc}{\lambda}$$

$$= \frac{(6.63\times10^{-34})(3\times10^{8})}{6.2\times10^{-12}}$$

$$= 3.21\times10^{-14}\,J$$

$$= \frac{3.21\times10^{-14}}{1.6\times10^{-19}}$$

$$= 0.201\,\text{MeV}$$

(ii) Compton scattering

(b) (i)

$$\frac{1}{E} - \frac{1}{E_0} = \frac{1}{m_e c^2}(1-\cos\theta)$$

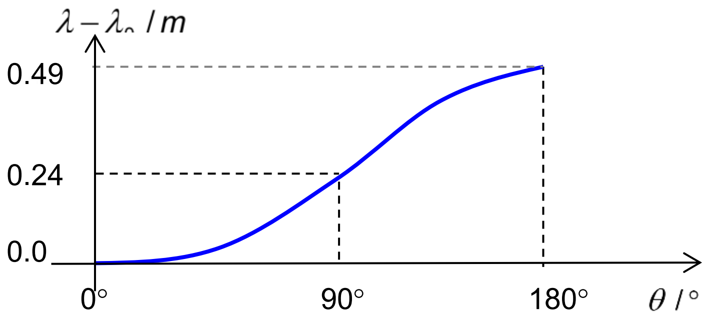
$$\frac{\lambda}{hc} - \frac{\lambda_0}{hc} = \frac{1}{m_e c^2}(1-\cos\theta)$$

$$\lambda - \lambda_0 = \frac{h}{m_e c}(1-\cos\theta)$$

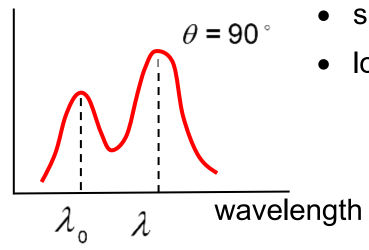
(ii) At largest Compton shift, $\lambda - \lambda_0 = \text{max}$

- $(1-\cos\theta) = \text{max} = 2,$
- $\cos\theta = -1$
- $\theta = 180^\circ$

(iii)



(iv) **1.**



- 2 peaks
- spacing $\lambda - \lambda_0$ between for 45° & 135°
- lower peak intensities

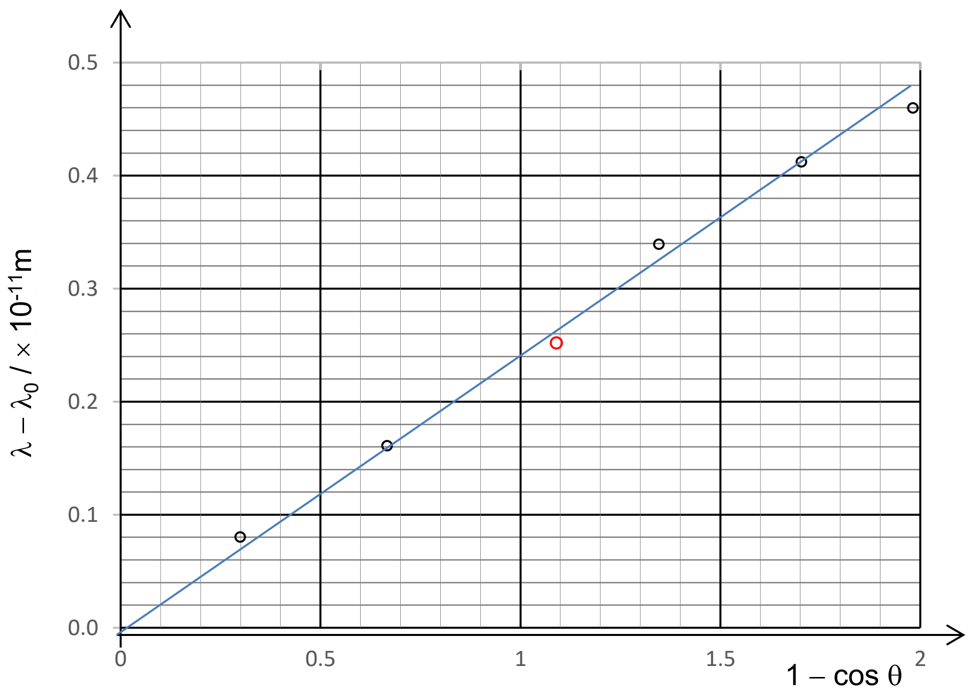
2.

- Only the incident photon is transmitted
- No scattered photon is produced

(v)

θ / °	45	70	95	110	135	160
$\lambda - \lambda_0$ / $\times 10^{-11}$ m	0.07	0.16	0.26	0.34	0.41	0.46
$1 - \cos\theta$	0.29	0.66	1.09	1.34	1.71	1.98

(vi)



1. Correct plot
2. Good balance of points
3. Correct read off of both coordinates

$$gradient = \frac{0.44 - 0.00}{1.80 - 0.00} = 0.244 \times 10^{-11} \text{ m} \qquad \frac{h}{m_e c} = 0.244 \times 10^{-11} \text{ m}$$

4. Correct read off when $1 - \cos\theta = 1.00$

From the graph,

$$\lambda - \lambda_0 = 0.24 \times 10^{-11} \text{ m}$$

$$\lambda = 0.24 \times 10^{-11} + 7.09 \times 10^{-11}$$

$$= 7.33 \times 10^{-11} \text{ m}$$

(c) (i) Energy produced = rest masses of 1 electron and 1 positron

$$E = 2\, m_e\, c^2$$

$$= 2\left(9.11 \times 10^{-31}\right)\left(3.00 \times 10^{8}\right)^2$$

$$= 1.6398 \times 10^{-13} \text{ J}$$

$$= 1.64 \times 10^{-13} \text{ J}$$

(ii)
- Higher energy will increase probability of the photon overcoming the field effect.
- Energy of photon should be larger as the electron and positron should be moving